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Research article

Integrating Environmental Management into the Digital Economy and Green Development of Enterprises: A Theoretical and Empirical Analysis

Shengnian Wang, Qianxing Jiang *, Jinke Wu

School of Economics and Management, Shihezi University, Shihezi, 832000, China

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ABSTRACT

Through the application of network platforms and digital technologies, the digital economy can help enterprises monitor and manage resource use in real time, optimize resource allocation, and improve resource utilization efficiency, thus reducing energy consumption and environmental impact. This paper uses the data of A-share listed enterprises in the Shanghai and Shenzhen stock exchanges from 2012 to 2023 to construct a panel data model and empirically tests the effect and mechanism of the digital economy on the green development of enterprises. The study finds that the digital economy can promote the green development of enterprises and can promote the green development of enterprises by stimulating green technology innovation and enhancing green management innovation. Further tests show that the promotion effect of the digital economy on corporate green development is more prominent in the higher degree of public environmental attention, market competition, executives with digital backgrounds, and enterprises belonging to the digital economy industry. The economic consequences test shows that the digital economy significantly improves corporate green total factor productivity by promoting corporate green development. The research conclusion provides a new idea for the implementation and planning of enterprise green development and provides an important reference for the government to formulate relevant digital economy and environmental protection policies.

1. Introduction

Amid the progressive advance of global digitalization, the urban digital economy has emerged as a vital driver of economic growth, innovation and development, and social progress (Wang et al., 2022). The development of digital economy is highly valued at home and abroad, and the Digital Economy Report 2024 has also been published by the United Nations Conference on Trade and Development across the globe. With the emphasis on the concept of the formation of the environmentally friendly, sustainable, and inclusive future digital environment, the report triggers the discussion of how the existing rapidly expanding digital economy affects the environment and the developing countries and the need to pursue sustainable development. It emphasizes the creation of greater value for developing nations while lowering the environmental input of the digital economy (Goldfarb and Tucker, 2019), and calls on the international community to cooperate in such a way that the digital economy will develop in a sustainable and inclusive way. Domestically, relevant departments issued the Work Priorities of Work for Digital Economy in 2024 and the Action Plan for Accelerating the Cultivation of Digital Talents to Support the Development of Digital Economy (2024–2026) to accelerate innovation and breakthroughs in digital technology and drive the digital economy to achieve high-quality development. China is bringing together all stakeholders, aligning with the overall trends in the development of the modern digital economy, and enjoying strong government support. It is also leveraging an array of technological tools and promoting collaboration to drive the development of China's digital economy.

In the modern environment of sustainable development in the world, green development in corporations is becoming one of the main concerns. Enterprise green development implies that the enterprises will be concerned with reduction of resource consumption, reduction of environmental pollution, and increase of ecological gains in their production and business operations to realize mutual benefits in both financial and environmental gains. This sustainable approach to business is not only beneficial to enable enterprises to preserve their long-term competitive advantages but also aligns with the constantly tightening regulations on the environment and the rising consumer awareness about environmental protection (Li et al., 2022). The European Commission published

* Corresponding author. Xinjiang Shihezi University: No.221 North Fourth Road, Shihezi City, Xinjiang, China.
E-mail addresses: wang-sn@shzu.edu.cn (S. Wang), qianxingjiang@163.com (Q. Jiang), 2385159009@qq.com (J. Wu).

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a programmatic document to focus on climate and environmental issues, the European Green Deal, in December 2019. It seeks to develop the EU into a prosperous and progressive society featuring an economically competitive, resource-optimized, and modern economy, and with the carbon footprint of greenhouse gases at zero levels in 2050 by making systemic changes to climate change, environmental risks, and forest and marine pollution. In September 2024, the Ministry of Ecology and Environment of China released a circular regarding measures taken by its Department for Ecology and Environment to further advance development of the private economy. The document clearly stated that the formulation and revision of pollutant emission standards should be accelerated, and the accounting standards and technical specifications for carbon emissions of enterprises in key industries and carbon emission reduction of projects need to be upgraded to steer the green and low-carbon advancement of companies. These are the reasons why international and domestic government agencies have issued a range of measures to vigorously advocate the green development of enterprises, mainly as follows: First, in terms of resource utilization efficiency, as core participants in economic activities, enterprises consume large volumes of energy and raw materials during their production and business operations. To attain green development means that enterprises need to streamline resource deployment, enhance resource use efficiency, and curb waste (Zheng et al., 2025), so as to ease the situation of resource shortage and ensure the sustainable operation of themselves and even the whole industrial chain. Secondly, in terms of environmental impact, the production behavior of enterprises is often accompanied by many environmental stressors, such as pollutant discharge and waste generation. Practicing this notion of green development and encouraging companies to actively adopt environmental protection technology and clean production processes can effectively reduce harmful effects upon this ecological environment and help maintain balance and restoration within this ecological system. Moreover, from the perspective of enterprises' own competitiveness, green development is no longer merely a reflection of social responsibility but also the key to winning market advantages (Shive and Forster, 2020). Consumers' preference for green products and services is increasing, and the policy level also tends to give more support and preferential treatment to green enterprises. Therefore, the green development of enterprises is not merely an unavoidable option conforming with the broader direction of the era but additionally an important strategic initiative linked to firms' own survival and long-term development. In addition, the current state pays more attention to environmental issues and the continuous enhancement of public consciousness of ecological conservation. With the continuous development of the digital economy, the traditional extensive enterprise development mode cannot be sustained, green development has emerged as a necessary path for enterprises to realize sustainable development.

In a word, the digital economy and green development, being the two key trends amid the current global economic and social transformation, are the indispensable pathways for China to attain its "dual carbon" goals. There is a close internal interdependence between the two, and their development trend and mutual correlation have become an important issue focused on by the industry and academia. The digital economy performs a multi-dimensional function in facilitating green development, which can not only improve the productivity of technical R&D and foster innovative cooperation(Xing et al., 2024) but also provide impetus for green technology innovation (Guo et al., 2023). It can also help enterprises monitor and optimize resource allocation in real time with the assistance of digital technology (Wang et al., 2021) to promote green management innovation; digital platforms can enhance the interaction and cooperation between enterprises and stakeholders (Xiao et al., 2023) and accelerate the transformation of green innovation achievements and the dissemination of environmental protection concepts. Meanwhile, as a typical representative of the new economic modality, the digital economy promotes the transition and modernization of conventional industries by virtue of frontier technologies including big-data analytics, cloud-based computing, the IoT, and blockchain; provides new development tools and platforms for enterprises; makes information circulation more efficient and operation more flexible; and injects new impetus into enterprises' green development (Zhao et al., 2022). These factors act in concert to boost the green development of firms and add new drivers to sustainable social development. China attaches considerable weight to the advancement of the two and continues to issue policies to promote coordinated progress in recent years. For example, in recent years, relevant departments have issued policy instruments, including the "14th Five-Year Plan for Digital Economy Development" and "Opinions on Accelerating the Comprehensive Green Transformation of Economic and Social Development." Against the backdrop of today's global sustainable development, examining the intrinsic connection linking urban digital economy and enterprise green development has profound theoretical value to steer enterprises onto the track of green and sustainable development and carries far-reaching significance for the overall sustainable development of society.

2. Literature review

2.1. Literature review on digital economy

When information technologies turn into a sphere in which the dynamic growth and mass consumption are also noticed, the digital economy turns out to be a potent force that can affect the economic situation worldwide. The digital economy is a new type of economic modality, it relies on digital technologies, takes data as a core factor input, and leverages contemporary information networks to enhance productivity (Kling and Lamb, 1999). In the past few years, the scientific community carried out the study of the profundity and holistic nature of how the creation of the digital economy influences the economy. With regard to the research perspective, the findings that are available are largely focused on two, i.e., macro and micro. At the macro level, being a novel type of economy, one of the most prevalent subjects of research has been its impact on macroeconomic functioning, policy-making, and industrial development. The digital economy can not only directly boost economic growth (Zhang et al., 2022) but also drive high-quality economic development (Ding et al., 2021; Ma and Wang, 2024; Shang et al., 2024), which can also trigger structural changes in employment(Qu and Fan, 2024) and have a damper effect on inflation (Sasongko et al., 2024). These impacts are far-reaching, and their breadth and depth fully demonstrate their core value as a new economic form in driving economic growth. As the flow of elements of data and empowerment of information is made effective, the impact of numerous policies, such as tax policies(Turina, 2020), monetary policies (Li et al., 2024; Wang, 2024), and even environmental policies (Xu et al., 2022; Xie et al., 2023), on the course of adjustment and the effectiveness of implementation in different spheres is noticeable, these policies contribute to the optimization of the resource allocation, and hence, they are effective in the industrial upgrading (Guan et al., 2022; Lu and Hu, 2024). It, in its turn, results in the high-quality formation of the industrial chain (Gao et al., 2024) and strengthens the resilience of the industrial chain (Hu et al., 2024). Moreover, these policies also play a positive part in energy transformation and changing of the environment (Xu et al., 2025). They also offer the much-needed support in the form of energy security and energy transformation (Dogan et al., 2025a). At the micro-level, the main focus of extant studies has been on the way the digital economy relates to and influences the three major themes of enterprise development, the competition in the marketplace, and consumers. The digital economy promotes innovativeness among enterprises via information technology and big data analysis, as well as the AI technologies (Zhao et al., 2022; Zhang et al., 2025), and contributes to the quality-based evolution of the enterprises (Liang and Tian, 2024; Ma and Wang, 2024), increases the extent of corporate risk-taking (Tian and Wang, 2014), and guarantees the increased foundation and support of long-term sustainable development of enterprises. The digital economy

has reshaped the nature of competition and laws of the enterprises, lowered the entry barrier of the market (Guo et al., 2022), and provided a more competitive environment to the market; a widespread adoption of digital technology has also increased the avenues of information acquisition to consumers (Birch and Adediji, 2025) and has also provided a more diverse resource through which corporate brands can be propagated (Küpper et al., 2021). In the meantime, the environment of the digital economy suggests that the transaction costs of the market grow exponentially due to the dissimilarity of consumer needs, and the introduction of digital technology reduces the cost of transaction (Nagle et al., 2020). In brief, Digital economy has produced significant economic consequences at the macro level, and also has an important impact on the micro level.

2.2. Literature review on corporate green development

The topic of sustainable environmental development has been of long-standing interest. In the context of this larger notion, corporate green development is one of the central elements, as the reflection and the actualization of environmental protection into the real world. The environmental development of the enterprise is determined by the achievement of the objectives connected with the sustainable production processes and minimization of harmful emissions. This is done by reducing energy and pollution at the same time, ensuring constant and healthy growth, a plan that will allow long-term sustainable development (Olson, 2008). Business, as the active participant in the economic processes, has a fundamental contribution to make in the movement of ecological development. The current academic studies concerning eco-friendly development of the business are usually based on the macro- and micro-perspectives of the problem. At the large-scale level, the corporate green developments are the practical way of applying the ideology of sustainable development. Such measures play a vital role in controlling and solving the international environmental issues. By implementing green technology and innovation, enterprises can curb carbon emissions and energy use, thus promoting environmental protection (Zhang and Li, 2020). The research shows that tax policy (Wang et al., 2024; Shi and Ge, 2025), the capital market (He et al., 2024), environmental regulation (Lu et al., 2025), and other factors all perform a pivotal function in advancing the green development of enterprises. Additionally, corporate green development can also bring broader socio-economic benefits, such as narrowing the income gap (Doğan et al., 2025), promoting gender equality (Fu et al., 2025), and driving innovation in renewable energy (Dogan et al., 2025b). The green development strategy guides the direction of the green development of enterprises, a factor that not only has far-reaching guiding significance with regard to the sustainable development of enterprises but also offers a guarantee for the comprehensive sustainable development of society (Wu et al., 2025). When implementing the Belt and Road Initiative, Wang and Wang (2023) noted that the governments are expected to play their guiding, promoting, and service roles. They ought to resolve the problems of lack of information services as well as the shortage of green development talent encountered by businesses that are involved in the Green Belt and Road Initiative construction through optimization of policy systems and enhancement of service support. This will result in the establishment of a good policy and talent atmosphere for the Green Belt and Road Initiative construction, enhance green development among the enterprises, and also help fulfill the global sustainable development goals. further advance the green development of firms, and support the fulfillment of global sustainable development goals. At the micro level, Afum et al. (2021) reported that enterprises should first reduce their costs through technological innovation, stimulate their willingness and motivation for green innovation, improve their operational efficiency, and achieve sustainable business goals by promoting environmental management and operation within the organization. At the same time, enterprises can cooperate with large online trading platforms to optimize resource allocation through big data to better meet customer needs, which will offer a better resource assurance underpinning the green development of enterprises (Gao et al., 2023). With the support of digital technology and platforms, companies can achieve green cooperation and good connection of resources with upstream and downstream suppliers and partners by building digital supply chain management systems, which can offer support for the green development of enterprises (Yin et al., 2020; Huang et al., 2021).

3. Theoretical analysis and research hypotheses

Being the core driving force of the emerging wave of science, technology revolution, and industrial transformation, the digital economy offers the new theoretical support and practical direction of green development of enterprises (Vial, 2019). Its core is to promote enterprises to achieve green transformation in resource utilization, production mode, and business model through data-driven, platform-enabling, intelligent leadership and integrated innovation (Si et al., 2024; Sun et al., 2024). Firstly, the urban digital economy helps enterprises accurately identify environmental risks, optimize resource allocation, and formulate scientific green development strategies through big data technology. At the same time, data analysis technology is used to carry out product life cycle assessment and promote green design, green production, and green supply chain management (Ge and Zhang, 2023). According to the resource-based theory, as a highly valuable strategic resource, digital technology assumes a central role in enterprises' development process (Simsek et al., 2024). Digital technology is able to help firms greatly enhance the effectiveness of resource utilization with its high efficiency and accuracy. Through scientific allocation and management of various resources, it can avoid unreasonable use and idle waste of inputs and advance green management innovation (Roh et al., 2022), so as to make steady progress on the road of sustainable development. It will be a significant advocate of green development (Li et al., 2023). The urban digital economy development will also facilitate the digital transformation of enterprises (Yoo et al., 2012), which can greatly enhance the dynamic capabilities of enterprises and cause them to demonstrate greater adaptability and responsiveness to environmental changes of various complexity and dynamic market needs. Digital transformation facilitates the better use of internal and external resources by enterprises with the assistance of new technologies and management ideas, the adaptability of the strategic positioning, and the readiness to face different challenges positively. Meanwhile, the advancement of the regional digital economy offers a powerful impetus towards the marketing of the green strategy of enterprises grounded in marketing the green management innovation of enterprises. By using the digital technology, businesses can conduct green research and development more efficiently, streamline the production process, cut down on energy usage, create the synergistic unity of green development and innovative development, and inject continuous vitality into the green development of enterprises (Quttainah and Ayadi, 2024).

Secondly, it is possible to rely on the constant innovation of green technologies by developing the regional digital economy aimed at the creation of an industrial Internet platform and an e-commerce platform. Green supply-side and demand-side innovation can become possible through the creation and upgrading of the platform, facilitate the R&D and use of green technology, lead green consumption, and increase the market of green products (Mu et al., 2022). On the one hand, artificial intelligence and intelligent manufacturing technology encourage transforming the enterprise production mode to intelligent and green to streamline production workflows and minimize energy consumption and pollution emissions. The digital economy has proven to be influential during the current era, according to the stakeholder theory. The digital economy, supported by cutting-edge information technology and an efficient communication platform, can greatly improve the transparency between enterprises and various stakeholders. Through digital channels, enterprises can more clearly show their operating conditions, development strategies, and fulfillment of social responsibilities to

shareholders, employees, customers, suppliers, and the public (Cheng and Zhang, 2023). On the other hand, the vigorous development of the urban digital economy has significantly strengthened the in-depth linkage and collaborative development between enterprises and upstream and downstream partners, industry organizations, regulatory authorities, and other stakeholders. All parties can conduct timely communication and feedback through the online platform and jointly take part in the policy-making and development procedure of enterprises, which enhances the scientific and democratic decision-making of enterprises (Wang and Wu, 2024). This increase in transparency and interactivity further enhances corporate social responsibility, enterprises attach greater importance to environmental protection, resource conservation, and sustainable development, actively fulfill their social responsibilities, and contribute to promoting green development (Wen, 2023). Driven by the digital economy, it can make enterprise products open and transparent from raw-material sourcing, manufacturing and treatment, sales of finished products, and other links, facilitate the fine management of enterprises and enhance consumers' environmental awareness through the introduction of terminal intelligence, guide consumers to carry out green consumption, promote the further research and development and production of enterprise green products, and drive the entire industry of enterprises in the market system to achieve green coordinated development.

Lastly, the proximity between the digital and the physical economies has also created new types of green businesses, such as the sharing economy and the circular economy. These provide the impetus for the increased green transition of conventional industries (Si et al., 2024). Nevertheless, the digital economy in the city continues to experience a set of challenges as it enables the development of businesses in a green way. They are data security risks, the problem of the digital gap, and the prohibitive expense of green technology. The innovation diffusion theory assumes that digital technology can be an essential factor in contemporary society because it is one of the innovations with significant revolutionary importance. Digital technology has the potential to spread in an enterprise within a short period due to its unique advantages and powerful capabilities. Having been implemented in a business, it might introduce an innovation to the customary working and management models, which boosts efficiency at work. This then minimizes the use of resources, hence creating a solid footing in encouraging green development (Wu et al., 2023). At the same time, it is also possible to disseminate digital technologies in industries. This will help various businesses to share ideas in order to push the implementation of green technologies and practices. The process of such kind can lead to the creation of digital platforms and networks, which would greatly drive the increase of green innovation. In the context of openness, interactivity, and high efficiency, it is possible that digital platforms can be used to announce the success of green innovation to both businesses and consumers on time to develop an interest and demand. The network of the internet enables the spreading of green innovation quickly to different areas and corporations, breaking the geographic and industrial barriers (Garad and Khalifa, 2024). Enterprises can also more easily acquire green technology information, gain the highest level of green practice experience, and enhance the level of green practice application through digital platforms and networks. This approach will not only guarantee that the companies are in a position to save on production and elevate the quality of products provided but also boost the likelihood of businesses competing and developing sustainably. Also, it will aid in motivating the industry as a whole to become involved in green development. In general, the digital economy offers a conceptual framework and realistic strategy of green development of businesses regarding technological growth and paradigm shift. It will become a constituent of the preconditions of the comprehensive green transformation of the economy and the society. Building on the preceding discussion, the hypotheses below shall be made in this paper.

H1. The development of the digital economy has significantly improved the level of green development of enterprises.

4. Research design

4.1. Sample selection and data sources

This paper takes China's Shanghai and Shenzhen A-share listed firms spanning 2012 to 2023 as the initial sample, excluding the missing values of ST, * ST, the financial industry, and main variables, and finally obtains 26,774 valid samples. Specifically, the digital inclusive finance dataset comes from the "Peking University Digital Inclusive Finance Index", and the remaining datasets are drawn from CSMAR, CNCA, the National Bureau of Statistics, CNRDS, and other databases, and all continuous variables are winsorized at the 1 % and 99 % levels.

4.2. Model construction and variable measurement

To examine the impact of the digital economy on the green development of enterprises, drawing on the studies by Wang et al. (2022) and Zhang et al. (2022), this paper develops the following model:

$$GREEN_{i,t} = \alpha_0 + \alpha_1 Dige_{i,t} + \alpha_2\, Control_{i,t} + \sum year + \sum ind + \sum province + \varepsilon_{i,t} \quad (1)$$

In which $DGREEN_{i,t}$ is the dependent variable of the current paper, which represents the green development of enterprises, and $Dige_{i,t}$ is the independent variable of the current paper, which represents the level of digital economy development of a city. In the independent variables of the present paper, the data is processed in such a way that the development level of urban digital economy is matched to the enterprise level i and t are the enterprise and the year respectively. When α_1 in the regression findings is highly positive, it means that the emergence of the digital economy will lead to green development of enterprises; if it is negative, it indicates that the digital economy will inhibit the green development of enterprises. In addition, referring to the research of Wang et al. (2022) and Zhang et al. (2022), we introduce control variables and control the possible impacts of year, industry, and province. Variable definitions are shown in Table 1.

4.2.1. Dependent variable

Enterprise green development. According to the findings of the research works by Ding and Cheng (2023), the paper constructs a system of assessment indices of enterprise green development that have three basic dimensions, namely economic profit, social value, and environmental benefits. The index of the comprehensive green development is determined with the assistance of the entropy method, and therefore the degree of green development of manufacturing enterprises is estimated quantitatively. Table 2 gives a description of the particular type of index selection and formulae applied in the calculation of the index.

4.2.2. Independent variable

Digital Economy. In relation to the publications by Zhao et al. (2020), the paper uses three indicators as the representatives of digital infrastructure, digital industries, and industrial digitalization. The entropy method and calculations are made and used to calculate the weights of each indicator that is used to make up a holistic regional digital economy development level indicator. Thereafter the digital economy indicators are matched to the enterprise level according to the registration place of the enterprise. The actual procedures of the digital economy indicator system choice and calculation are presented in Table 3.

4.2.3. Mediating variables

Green technology innovation. Drawing on the work of Jia and Jing-bo (2020), this study adds 1 to the total count of green invention

Table 1
Definition of variables.

Type of variable	Name of variable	Symbol of a variable	Description of variables
Dependent variable	Green development level of enterprises	GREEN	The comprehensive index system of enterprise green development is constructed from the three dimensions of economic profit, social value, and environmental benefit, and the entropy method is used to calculate the comprehensive index of green development to measure the degree of enterprise green development.
Independent variables	The Digital Economy	Dige	Using the three representative indicators of digital infrastructure, digital industrialization, and industrial digitalization, the entropy method is used to determine the weight of each indicator and calculate it, and finally a regional comprehensive index of provincial digital economy development level is synthesized. Then, the listed companies are matched according to the addresses registered in the provinces where they are located, and the corresponding enterprise data are obtained.
Mediating variable	Green technology innovation	EnvrPat	The sum of the number of green invention patent applications and the number of green utility model applications is added by 1, and then the natural logarithm is taken.
	Green management innovation	GM	The CRMAR environmental database stores the publicly available data on the listed companies in respect of environmental supervision and environmental certification, which is the ISO14001 environmental management system certification and ISO9001 quality management system certification. Meanwhile, the records of the management disclosure of the listed companies are aggregated at the same time as the environmental protection management system, environmental protection education and training, and special environmental protection actions. Equalization of the above five indicators and then adding them up gives the total score, and it constitutes the proxy variable of the level of corporate green management innovation (GM).
Moderating variable	Public environmental concern	PEC	The original data of environmental attention of nationally listed companies and official accounts are matched to obtain the daily average data of the Baidu search index of listed companies.
	Degree of market competition	HHI	The Herfindahl index
	Executives' digital background	Ceodig	In case any senior executive of a listed company holds a digital background, the value is 1; otherwise, it is 0.
	Digital economy industry	Digeind	Guidelines on Industry Classification of Listed Companies of 2012 include computer, communication, and other electronic equipment manufacturing; software and information technology services; Internet and related services; telecommunications; and radio, television, and satellite transmission services as digital economy industries.
Control variables at the micro level	Size of enterprise	Size	The total assets of the firm are taken as the natural logarithm.
	Asset-liability ratio	Lev	Liabilities/Total assets.
	Return on assets	ROA	Profit before deductions/ amount of assets
	Operating income growth rate	Growth	Growth rate of sales revenue of the company
	Size of the Board	Board	Total number of directors
	Dual roles	Dual	Assign a value of 1 if the two positions are held concurrently, and 0 otherwise.
	Concentration of ownership	TOP1	Shareholding ratio of the largest shareholder
	Nature of property rights	SOE	A dummy variable that equals 1 when the company is state-controlled and 0 when the company is privately owned.
Macro-level control variables	Level of transport infrastructure	Road	The logarithm of highway mileage is taken.
	Level of informatization	Infor	Total post and telecommunications business divided by regional GDP
	Degree of government intervention	Gove	Fiscal expenditure divided by gross regional product
Other control variables	Year dummy variable	Year	Control for year effects
	Industry dummy variable	Ind	Control for industry effects
	Dummy variable of province	Province	Controlling for the effect of provinces

patent filings and green utility model filings and then takes the natural logarithm (EnvrPat) as the index to measure GTI.

Green Management Innovation. The data used in the present paper are the disclosures of the listed companies. The information generated in the CSMAR environmental database is related to the supervision and certification of the environment. These are certifications under the ISO14001 environmental management standard and the ISO9001 quality management standard (Zhao et al., 2015). Further, the disclosures of the environmental protection management of the listed companies are also combined, i.e., the environmental protection management systems, environmental protection education, training, and records of the special action for environmental protection measures

Table 2
Comprehensive index system of enterprise green development.

Name of variable	Type of dimension	Type of indicator	Specific calculation method
Green development of enterprises	Economic profit	Net profit margin on total assets (+)	The total corporate net profit average of the total corporate assets ratio.
		Net profit growth rate (+)	The net profit of the company in this period has been up compared to the last period.
		Inventory to revenue ratio (+)	Inventory/operating income
		Net fixed assets (+)	The cumulative depreciation, followed by the impairment provisions of a fixed asset taken away, and the original value of the asset.
		Total factor productivity (+)	TFP is measured by the use of the fixed effect method (FE).
		Size of enterprise	Logarithm of total assets of the firm
		Operating costs (−)	Cost of principal operations + Cost of other operations
		Selling expenses (−)	Add up all sales expenses.
		Administrative expenses (−)	Other expenses incurred in the administrative section of the venture to administer and plan the business operations include company funds, labor union funds, education funds of the employees and the laborers, labor insurance premiums, unemployment insurance premiums, director dues, consultation fees, and audit fees.
	Value of society	Earnings per share (+)	Earnings per share = (net profit during the period and less preference dividends)/ annual weighted average total share capital
		Remuneration paid to employees (+)	Total compensation paid to employees
		Number of employees (+)	Logarithm of the number of employees in nature.
	Environmental benefits	Environmental tax (+)	The logarithm of the main business income/ natural logarithm of environmental taxes.
		Whether passed ISO9001 certification (+)	If the enterprise has passed ISO9001 certification, the value is 1; otherwise, it is 0.

in text form, and the scores of all five indicators are summed up by equal weight. A proxy variable represents the degree of corporate green management innovation (GM), which is formed on the basis of the overall score received.

5. Analysis of empirical results

5.1. Descriptive statistics

Apparently, this research article has the descriptive statistics which are indicated in Table 4. The following are the statistical characteristics of the digital economy. The mini and max are 0.0166 and 0.7466 respectively, the average is 0.2662 and the standard deviation is 0.1727 in the digital economy. The value of 0.2662 is an average value based on the distribution pattern of the present data, it is an indication that the level of development of the digital economy of the individual province can be enhanced. An economy with a high level of digital economy can only be created when most of the provinces achieve incredible growth. A standard deviation of 0.1727 indicates a considerable dispersion in the development of the digital economy across provinces, it means that the growth of the digital economy is not in equilibrium with the corresponding level of all enterprises in the various provinces. Thus, a disparity exists in the level of development of various enterprises in the digital economy. In case of Corporate Green Development, the lowest value is 0.0042, highest value is 0.4618 and the average is 0.0398 with a standard deviation of 0.0533. The analysis of such information allows us to conclude that there is some gap in the sphere of corporate-level green development. Such extremes of 0.4618 and 0.0042 show that not every enterprise among the sample has yet achieved a high level of green development and might be yet unable to develop effective practices in such areas as energy saving and emission reduction, resource recycling, and green supply chain management. Enterprises that have reached the higher level can benefit from technological invention and management optimization, thereby achieving a win-win situation of economic and environmental benefits. The average of 0.0398 shows that overall level of the green development of the sample enterprises needs further enhancement, most of the businesses still have far to go before they can attain the high level of green development. The standard deviation of 0.0533 indicates that the dispersion degree of enterprises' green development level is relatively moderate, it signifies that there are various prerequisites at which a corporate green development is conditional upon such aspects as the characteristics of the industry, the strategic location of the corporation, the constraints of the policy, and the effectiveness of incentives. The rest of the control variables are also distributed fairly.

5.2. Regression analysis

To test the hypothesis of this paper, which is whether the development level of the digital economy influences the green development of enterprises, a regression test was initially carried out on model (1). Table 5 reports the outcomes of it. Column (1) will be the simple regression with no control variables that will indicate the effect of the digital economy on the green development of the enterprises. The coefficient of 0.0166 is extremely positive at 1 %. When control variables are used, the regression results are pointing to column (2). Dige value of 0.0121 is rather positive at the 5 % level, which implies the more the development level of the digital economy improves, the more the green development of enterprises, thereby fulfilling hypothesis H1 of the given paper.

5.3. Robustness test

After in-depth research and analysis, so as to guarantee the reliability and validity of the conclusions obtained, the robustness test is a crucial link. The robustness test helps us assess whether the results are stable for different assumptions, model settings, and data processing methods. Through testing, we can enhance our confidence in empirical conclusions and provide a more solid foundation for subsequent theoretical discussion and practical application. Next, we will perform a robustness test for these research findings of the present paper to further verify their

Table 3
Indicator system of digital economy.

First-level indicators	Second-level indicators	Third-level indicators	Specific definition of indicators	attribute
The Digital Economy	Digital infrastructure	Internet broadband access rate	Internet broadband for seaports/permanent residents in the area.	+
		Internet broadband penetration rate	Internet broadband access clients/permanent residents in the area	+
		Scale of mobile phone facilities	Mobile phone switch capacity	+
		Length of long-distance optical cable line	Length of long-distance optical cable line	+
		Number of web pages	Direct data	+
		Number of domain names	Direct data	+
	Industrialization of digital	Total amount of telecommunications services per capita	Total amount of telecom services/area of permanent residence Number of population	+
		Mobile phone penetration rate	Direct data	+
		Legal entities in the information transmission, software, and information technology services.	Direct data	+
		Proportion of employment in information software industry	Information technology services, software, and information transmission Employment in urban units/employment in urban units	+
		Domestic number of patent applications granted	Direct data	+
		Count of domestic patent applications to be accepted	Direct data	+
	Digitalization of industries	Peking University Digital Inclusive Financial Index	Direct data	+
		Share of enterprises involved in e-commerce transactions	Direct data	+
		E-commerce sales	Direct data	+
		Websites per hundred enterprises	Direct data	+
		Added value of secondary and tertiary industries	Added value of the secondary industry + added value of the tertiary industry	+
		Investment in technological and scientific research	Industrial enterprises spending above a specific size on R&D	+
		Express delivery volume	Direct data	+

Table 4
Descriptive statistics of main variables.

Variable	Obs	Mean	SD	Median	Min	Max
GREEN	26,774	0.0398	0.0533	0.0081	0.0042	0.4618
Dige	26,774	0.2662	0.1727	0.2278	0.0166	0.7466
Size	26,774	22.3190	1.2946	22.1319	19.5245	26.4523
Lev	26,774	0.4246	0.2018	0.4155	0.0381	0.9077
ROA	26,774	0.0343	0.0656	0.0351	−0.5562	0.2223
Growth	26,774	0.3617	0.9891	0.1296	−0.9284	12.1957
Board	26,774	2.1184	0.1963	2.1972	1.6094	2.7081
Dual	26,774	0.2800	0.4490	0.0000	0.0000	1.0000
Top1	26,774	0.3406	0.1466	0.3177	0.0780	0.7579
SOE	26,774	0.3725	0.4835	0.0000	0.0000	1.0000
Road	26,774	11.9566	0.9156	12.2122	7.0273	12.9815
Infor	26,774	0.0466	0.0340	0.0381	0.0026	0.2356
Gove	26,774	0.1969	0.0976	0.1798	0.1050	1.3538

applicability and stability in different situations.

5.3.1. *Replacing independent variable*

Following the methodology of Xinwu et al. (2022), the word frequency (Dec) associated with the digital economy, including the terms "digital economy" and "digitalization," from the Baidu Index across 31 provinces and 295 prefecture-level cities is crawled and collated via R language for gauging the regional attention toward the digital economy in each province. Having matched on the basis of the registered location of the company, the particular regression values are represented in column (1) under Table 6. The digital economy difference with respect to the 10 percent level of the green development of enterprises is a significantly positive one, confirming the hypothesis of this paper.

5.3.2. *Modify the sample interval*

This study originally selected data spanning 2012 to 2023 as the empirical sample. To be more precise, the appearance of the COVID-19 pandemic at the end of 2019 not only impacted the macro-policy environment but also the business environment of enterprises to a considerable extent. The alterations in the policy and the market provoked by the epidemic can disrupt the normal flow of the corporate social responsibility implementation and, therefore, the internal logic between the variables of the research. To strengthen the validity of the study results, a modification of the sample time frame is done in this section. It does not take into account the values of the years 2020 and 2023, cogitates about the values preceding the epidemic, and re-estimates the effect stemming from the formation of the digital economy on the green development of the enterprises. These are the outcomes as indicated in Table 6 in column (2). The regression findings demonstrate that there is a positive yet significant value of the impact of the digital economy on the green development of corporations at the 1 % level of significance. This validates the hypothesis of this paper.

Table 5
Basic regression results.

	(1)	(2)
	GREEN	GREEN
Dige	0.0166***	0.0121**
	(2.976)	(2.092)
Size		0.0036***
		(5.131)
Lev		0.0036
		(1.243)
ROA		0.0039
		(0.801)
Growth		−0.0002
		(-0.818)
Board		−0.0076***
		(-2.910)
Dual		−0.0008
		(-0.858)
Top1		−0.0019
		(-0.397)
SOE		0.0037**
		(1.995)
Road		−0.0035
		(-1.515)
Infor		−0.0106
		(-0.543)
Gove		0.0042
		(0.283)
_cons	0.0354***	0.0113
	(23.488)	(0.357)
N	26,774	26,774
Adj. R^2	0.4916	0.4925

Table 6
Robustness test.

	(1)	(2)	(3)	(4)
	GREEN	GREEN	GREEN	GTFP
Dec	0.0122*			
	(1.936)			
Dige		0.0323***		
		(3.309)		
DID			0.0032**	
			(2.547)	
GREEN				0.0050*
				(1.692)
Size	0.0036***	0.0013	0.0036***	−0.0004
	(5.149)	(1.451)	(5.173)	(-1.246)
Lev	0.0037	0.0036	0.0039	0.0011
	(1.282)	(0.968)	(1.376)	(0.890)
ROA	0.0039	−0.0015	0.0039	−0.0019
	(0.784)	(-0.241)	(0.782)	(-0.891)
Growth	−0.0002	−0.0003	−0.0002	−0.0001
	(-0.818)	(-1.263)	(-0.797)	(-0.508)
Board	−0.0076***	−0.0059*	−0.0076***	−0.0009
	(-2.929)	(-1.752)	(-2.942)	(-0.825)
Dual	−0.0008	−0.0018	−0.0008	−0.0006
	(-0.867)	(-1.477)	(-0.859)	(-1.513)
Top1	−0.0019	−0.0023	−0.0026	−0.0002
	(-0.407)	(-0.380)	(-0.546)	(-0.090)
SOE	0.0038**	0.0033	0.0039**	−0.0003
	(2.015)	(1.106)	(2.091)	(-0.333)
Road	−0.0032	−0.0023	−0.0041*	0.0008
	(-1.378)	(-0.729)	(-1.800)	(0.816)
Infor	−0.0076	−0.0049	−0.0155	0.0187**
	(-0.385)	(-0.195)	(-0.788)	(2.029)
Gove	0.0048	−0.0031	0.0126	0.0043
	(0.317)	(-0.132)	(0.860)	(0.601)
_cons	0.0077	0.0404	0.0194	1.0365***
	(0.242)	(0.935)	(0.619)	(73.166)
N	26,774	16,435	26,774	26,774
Adj. R^2	0.4925	0.5168	0.4925	0.9678

5.3.3. Quasi-natural experiment based on "Gigabit urban construction demonstration area"

To empirically assess the micro-level mechanism via which the extent of digital economy development influences enterprise green development, this study utilizes the "Gigabit City" policy as a quasi-natural experiment. Assign a score of 1 when a city is designated as a Gigabit City during the current year or in any later year, and assign 0 otherwise. It will be grounded in the use of the Difference-in-Differences (DID) estimation tool, which we will use to evaluate the micro-level effect mechanism of the digital economy establishment on corporate green development and form the following DID model:

$$GREEN_{i,t} = \alpha_0 + \alpha_1 DiD_{i,t} + \alpha_2\, Control_{i,t} + \sum stkcd + \sum year + \sum province + \varepsilon_{i,t} \quad (2)$$

In which the dependent variable $GREEN_{i,t}$ is the level of green development of the corporations of enterprise i in the year t, the fundamental independent variable, $DiD_{i,t}$ is whether the city where enterprise i is situated is listed as a "gigabit city" in year t, in order to capture the digital economy development policy. The control variables are denoted by $Control_{i,t}$, the fixed effect of the individual is denoted by stkcd, the time fixed effect is denoted by year, the province fixed effect is denoted by province and the random error term is denoted by $\varepsilon_{i,t}$ The regression findings are indicated in column (3) of Table 6, and the findings of the regression of gigabit cities on the green development of the enterprises are significant at 5 % level, thus supporting the strength of the research findings of the present study.

5.3.4. Instrumental variable test

Choosing suitable instrumental variables for core independent variables constitutes the primary approach to addressing the endogeneity problem. Following the approach proposed by Nunn (2014), the interaction term between the 1984 per 100 capita fixed-line telephone penetration and the prior year's number of Internet subscribers is used as the instrumental variable. The justification is outlined below: First, the interaction term is significantly related to the digital economy.The number of fixed telephones in 1984 indicates the presence of early communication infrastructure in the area that offers the hardware base upon which the creation of the digital economy may occur. The number of internet users that was registered in the previous year is an indicator of the infiltration of digital applications and demand-ecology. The interaction term between the two can adequately explain the development potential of the digital economy and has a strong relationship with the development stage of the digital economy. Second, the cross-term of interaction satisfies the exclusivity constraint. Fixed telephone subscriptions in 1984 were determined by historical communication policy and population density, which do not have a direct link to the current green decision of enterprises. The prior year's growth in Internet subscribers depended on the popularization of digital technology and consumption habits, not on the reverse demand of enterprises for green, and without other influence paths outside the digital economy. Based on the research by Huang and Chen (2023), the two-stage least squares algorithm (2SLS) is used to construct the estimations related to the instrumental variables, and the results concerning the particular findings are presented in Table 7. The regression outcomes indicate that the regression coefficient of the digital economy on corporate green development is positive and passes the significance test, which further verifies the core conclusion indicating that digital economy development can significantly promote corporate green development. Regarding the verifying the validity of instrumental variables, the test results of LM statistics show that there is no occurrence of instrumental variable underidentification; the value of Wald F statistics is higher than the critical value of 16.38 at the significance level of 10 %, so it is feasible to judge that there exists no issue of weak instrumental variables. Based on the test results above, it can be concluded that the research conclusion of the digital economy promoting the green development of enterprises has strong robustness.

6. Further research

6.1. Mechanism analysis

Nowadays, the nation attaches substantial importance to the green development of enterprises. The country has steadily rolled out targeted policies to propel green development among businesses. Simultaneously, the continuous development of the digital economy has served as a key driver in further facilitating green development. However, what constitutes the bridge between the digital economy and green enterprise development, and what factors play an intermediary role in this process, are core issues that urgently require in-depth exploration and discussion. The following will examine and discuss these two intermediaries: green technology innovation and green management innovation.

Table 7
Instrumental variable test.

	(1) One-stage regression: Dige	(2) Two-stage regression: GREEN
Dige		0.0683***(2.84)
IV	0.0006***(28.77)	
controls	Controls	Controls
stkcd/province/year FE	Controls	Controls
Kleibergen-Paap rk LM		1126.915***
Kleibergen-Paap rk Wald		827.668
N	26,564	26,564

6.1.1. Green technology innovation

The enterprises need to be under the support of green technologies to attain green development (Guo et al., 2023). The evolution of the digital economy may trigger governments to invest more in digital infrastructure (Liang and Li, 2023) to support green innovation of enterprises, offering them technical support and guaranteed resources. The large-scale use of big data analytics and artificial intelligence technologies with the evolution of the digital economy allows the enterprises to be more innovative in green technologies, optimize the production process, cut waste of resources, and support the further evolution of the enterprise. To begin with, as the digital economy is developed, governments will invest more in the creation of digital infrastructure. Digital infrastructure assumes the core roles of information transmission, processing, and storage in the modern socio-economic system and is one of the main pillars that leads to the orderly work of the digital society and its sustainable development. Being an indispensable part of the age of the digital economy, digital infrastructure creates an inter-network interface of enterprises and personalities by creating a stable and high-speed network architecture, enhancing the efficiency of information interaction and resource integration capabilities of social actors to a large extent. This is a good base for the green technology innovation of enterprises, and green technologies can significantly stimulate the green development of enterprises. Secondly, digital technologies boost the effectiveness of information gathering, processing, and dissemination, contributing to the visualization and transparency of information data and a spillover effect, which allows disseminating information as widely and extensively as possible and using technological capabilities (Zhang et al., 2022). Regarding the visualization aspect, businesses will be able to understand more true information about the real-time data on the consumption of resources, the creation of waste, and the environmental impact and further stimulate the implementation and development of green technologies. On transparency, more information transparency also allows consumers and investors to access more corporate environmental information (Zakaria et al., 2021), which encourages enterprises that are more actively involved in green innovation to gain a reputation in the market, external support, and incentives them to invest more in this area, thereby facilitating green change in corporations. As far as the spillover effect is concerned, digital technologies facilitate information exchange and cooperation across organizational boundaries and lower the cost of technology adoption for companies, as well as increase the capacity to collaboratively innovate by sharing the technological achievements and innovative experience (Xing et al., 2024), which facilitates the joint process of green development of the enterprises. Lastly, governments and regulatory authority can use digital technologies to monitor and manage the environment through environmental policing. By using the corresponding software and digital technologies, it is possible to monitor the pollution situation of enterprises faster and more precisely and make enterprises improve and upgrade their technologies. The stringent environmental regulations have posed pressure on business to comply with the environmental protection laws of environmental protection and as a result, promote the development of green technology. According to Jia and Jing-bo (2020), innovation of green technology in corporate is calculated as (number of green invention applications of patents plus number of green utility model applications) +1 and then the natural logarithm is taken (EnvrPat). The results are indicated in column (1) of Table 8. The interaction term between the digital economy and the corporate green technology innovation is very significant at 1 % level which verifies the mediating effect of corporate green technology innovation variable.

6.1.2. Green management innovation

The digital economy is booming at present, posing new opportunities and challenges to businesses on a scale never seen before. Concurrently, the focus on environmental protection on the global level is continuously growing, making green development an unavoidable pattern in the context of enterprises (Jiang and Lyu, 2022). The bridging aspect between green enterprise development and the digital economy is played by green management innovation. To start with, the digital economy helps businesses to gather and process vast volumes of data on the environment and operations. Enterprises can keep track of their essential indicators like energy consumption, waste emissions, and resource usage in real-time by virtue of technologies like sensors and the Internet of Things (IoT) (Zhu and Lu, 2024). This information not only serves the purpose of offering precise information upon which the enterprises can develop green management strategies and thus help them in identifying the possible issues and areas of improvement in their environmental condition, but also, by analyzing the enterprise and market data, enterprises are also able to determine their current position and their future growth opportunities, hence producing better green strategic plans and offering a guide to their green developments. Secondly, the emergence of the digital economy creates fresh opportunities for the digital transformation of supply chain management, which strongly advances the green management practice deepening in enterprises, companies use digital supply chain solutions to evaluate and oversee the eco-efficiency of their suppliers. The extensive use of digital platforms and communication technologies will considerably enhance the effectiveness of information exchange between all actors in the supply chain to support real-time communication and cooperative interactions and allow all parties to develop green supply chain objectives and plans together (Feng et al., 2022). Meanwhile, through the assistance of big data, the Internet of Things, and other digital technologies, businesses can track and precisely monitor the logistics, information flow, and capital flow of each node of the supply chain in real time and enhance the visibility of supply chains and optimize their decisions in this regard (Wan et al., 2022). Encouraging suppliers to enter into partnership with suppliers who have a good environmental management

Table 8
Mediation test.

	(1)	(2)
	GREEN	GREEN
Dige	0.0084	0.0199***
	(1.431)	(4.461)
EnvrPat	−0.0023**	
	(-2.138)	
Dige × EnvrPat	0.0084***	
	(2.763)	
GM		0.0259***
		(53.007)
Dige × GM		0.0034**
		(2.393)
Size	0.0035***	−0.0022***
	(5.006)	(-3.827)
Lev	0.0035	0.0046**
	(1.222)	(2.058)
ROA	0.0039	−0.0011
	(0.786)	(-0.292)
Growth	−0.0002	0.0000
	(-0.832)	(0.017)
Board	−0.0074***	−0.0045**
	(-2.855)	(-2.218)
Dual	−0.0007	0.0000
	(-0.821)	(0.018)
Top1	−0.0018	−0.0078**
	(-0.371)	(-2.071)
SOE	0.0036*	0.0000
	(1.935)	(0.025)
Road	−0.0034	−0.0017
	(-1.476)	(-0.932)
Infor	−0.0108	−0.0136
	(-0.552)	(-0.867)
Gove	0.0043	0.0052
	(0.289)	(0.420)
_cons	0.0128	0.0834***
	(0.407)	(3.305)
N	26,774	26,774
Adj. R^2	0.4927	0.6851

history will guarantee green sourcing of raw materials and the development of a complete green supply chain system that minimizes the chances of environmental risks in the supply chain and provide the firm with a strong base of sustainable development. Last but not least, the green corporate culture can be created with the help of green performance evaluation, green human resource management, and respective rules and regulations, allowing enterprises to operate, manage, and develop within the framework of green ecology. This unites the business operations and daily activities of the enterprise with the philosophy on green development, and it has a green form of management that is consistent with the enterprise plans. Building upon the foregoing analysis and drawing on the research of Zhao et al. (2015), this paper converts green management innovation into a five-dimensional index system, and takes the ISO14001 and ISO9001 certification status disclosed by listed companies in the sub-database of "Environmental Supervision and Certification Disclosure" of CSMAR environmental database. The presence or not of the three institutional practices of environmental protection system, environmental protection education and training, and environmental protection special action in the sub-database of "management disclosure" are superimposed, and the binary assignment and summation are carried out, and the comprehensive score is finally constructed as the proxy variable of enterprise green management innovation (GM). As shown in column (2) of Table 8, the interaction term of digital economy and enterprise green management innovation is significantly positive at the level of 5 %, which verifies the mediating role of enterprise green management innovation.

6.2. Heterogeneity analysis

6.2.1. Public environmental attention

As the economy of my country is growing steadily and continuously and people are living in a more comfortable and better condition, the population has become more concerned with environmental protection, and the issues connected with it have become the main concern of the population. It is against this background that an urgent, intimate, intrinsic tie has formed between the digital economy and green development of corporations, where public environmental awareness is playing a key regulatory role in the course of the digital economy spawning corporate green development. On the one hand, corporate green development can be facilitated with the assistance of higher environmental awareness of the population (Li et al., 2023). In the digital economy era, information dissemination is convenient and rapid. The general population can more easily access the information connected with the environmental behavior of the corporations. When people are highly concerned about the environmental issues, to preserve their image and market share, companies need to address such issues as green production and green management (Chen, 2022). Companies under such pressure from the populace will invest more into environmental protection and implement a row of policies to obtain green development, like saving energy, cutting down emissions, and adopting cleaner production. Simultaneously, the more companies are responsive to the idea of green development and attain high environmental outcomes with the assistance of the digital economy, the more socially recognizable and supported they will become. This does not only assist the companies in drawing investment but also increases market share, thereby generating a virtuous cycle in the development of enterprises. Conversely, the digital economy is developed through the guiding influence of public environmental awareness. As more people become environmentally aware, they will buy green and environmentally friendly products and services. Digital economy enterprises will constantly innovate and optimize green technologies to satisfy this need on the market, and the industrial chain will push towards green and low-carbon development (Xue et al., 2022). This will entail environmental awareness among people, where the corporate social responsibility of digital economy enterprises will be a priority, thus generating a win-win pattern of both economic and ecological benefits; that is, the ecological benefits create a promoting impact on the digital economy and green development of enterprises. The gauging system of the environmental awareness of people in this paper was developed using a quantitative index of the search index and information index of "smog" and "environmental pollution" using the Baidu platform of different prefecture-level cities and provinces. This system involves the evaluation of the level of public interest in environmental problems in the various regions and classifies the information according to the median. The findings are exhibited in columns (1) and (2) of Table 9. As is evident in column (1), this result is significant at the 5 % level when the environmental awareness of the public on the listed companies is also high; hence, when the environmental awareness of the public on the listed companies is low, the result is not significant, as can be seen in column (2). This finding facilitates the study in this paper, i.e., that a greater degree of public environmental awareness is positively moderating in the phenomenon of the digital economy leading to corporate green development and can reinforce the driving influence of the digital economy on corporate green development.

6.2.2. Marketization level

Marketization is an indicator of how much market forces contribute to the distribution of resources in an area. The increased level of marketization means that the market will have a leading function in resource allocation, and the resources are going to be allocated to the market in a better way. In that regard, the influence of the digital economy in facilitating the green development of enterprises might be more prominent in a situation where marketization is at a high level. The reason is that in a highly market-oriented environment, enterprises experience closer competition, which compels them to use digital technologies to save prices and enhance efficiency (Kim et al., 2020), which will ensure their green development and a competitive advantage. At the same time, market mechanisms are able to better transfer the signals and values of green development, which will enable enterprises to gain clearer insights into consumer demand for green products and therefore invest more in green development, leveraging the digital

Table 9
Heterogeneity analysis of public environmental attention and marketization level.

	(1)	(2)	(3)	(4)
	GREEN	GREEN	GREEN	GREEN
Dige	0.019**	0.011	0.008	0.020**
	(2.22)	(1.03)	(0.78)	(2.33)
Size	0.004***	0.004***	0.002**	0.005***
	(4.23)	(2.63)	(2.26)	(4.67)
Lev	−0.003	0.003	0.005	0.003
	(-0.61)	(0.52)	(1.09)	(0.59)
ROA	−0.006	0.010	0.010	−0.008
	(-0.74)	(1.39)	(1.40)	(-0.95)
Growth	−0.000	−0.000	−0.000	−0.000
	(-1.00)	(-0.15)	(-0.89)	(-0.03)
Board	−0.006*	−0.010**	−0.003	−0.010***
	(-1.65)	(-2.23)	(-0.87)	(-2.63)
Dual	−0.000	−0.002	−0.001	−0.002
	(-0.09)	(-1.26)	(-0.79)	(-1.54)
Top1	−0.006	0.014	−0.005	0.003
	(-0.84)	(1.57)	(-0.61)	(0.46)
SOE	0.003	0.004	0.003	0.005
	(1.18)	(1.26)	(1.17)	(1.60)
Road	−0.006*	−0.000	−0.003	−0.003
	(-1.83)	(-0.09)	(-0.78)	(-0.93)
Infor	−0.019	−0.008	−0.006	−0.025
	(-0.53)	(-0.33)	(-0.21)	(-0.83)
Gove	0.009	−0.012	−0.004	0.010
	(0.40)	(-0.50)	(-0.16)	(0.49)
_cons	0.024	−0.020	0.025	−0.028
	(0.54)	(-0.34)	(0.51)	(-0.59)
N	13,218	12,992	13,297	13,477
Adj. R^2	0.459	0.555	0.517	0.503

economy. On the other hand, in a situation where marketization is low, the role of the digital economy might be somewhat restricted. Since the market does not operate perfectly and some barriers to the movement of resources exist, businesses will be less susceptible to market forces that promote green development and will not have the motivation to use digital technologies to promote green transformation. Besides, the administrative intrusion and barriers to the market may be more obtrusive, so it might be hard to realize the full potential of the digital economy in the development of green areas. This study draws on the research of Peress (2010), and the Herfindahl Index (HHI) is one of the essential measurement indicators. The HHI index has a marked positive association with the market concentration; namely, the greater the HHI index, the higher the concentration of the market share in the hands of a few operators and the less the market competition. On the other hand, a low HHI index implies that the market competition is stronger. Median-grouped regression analysis of the HHI index yielded the findings presented in columns (3) and (4) of Table 9. According to column (3), when the HHI index is high, the market competition level remains low, and the outcome is not substantial. Column (4) reveals that as the HHI index decreases, the market's competitiveness level turns out to be so high that the outcome achieves statistical significance at the 5 % significance level. This demonstrates that the more market competition is elevated, the more the positive promoting impact of the digital economy advancement level on the green development of businesses becomes more apparent.

6.2.3. Digital background of senior executives

The digital background of the executives is the knowledge, skills, and experience of executives in both information technology and in digital finance that holistically encompasses the professional educational history and work experience of the executives in the digital arena. In the modern world of active growth of the digital economy, the digital background of executives is extremely important in terms of green development of enterprises. It is based on this that the current paper examines the issues present in the digital background of executives and proposes the relevant countermeasures. With their high level of professionalism and extensive experience, in general, the senior executive associates of digital backgrounds tend to be more sensitive to the trends and opportunities of the digital economic development. First, in terms of the digital aspect of resource acquisition, digital-based executives are better acquainted with how the digital economy operates, and according to the law of development, they can better reflect on the new resources introduced by the digital economy with their incisive analysis. Due to the large endowment of the digital economy, new types of resources and access channels persist, and these executives can quickly discover and absorb these resources to guarantee a solid resource base for the green development of enterprises, ensuring that enterprises take an advantageous place in the process of green transformation. Second, regarding decision-making, the digital background of executives provides a special advantage to them, as they will better understand the ability of the digital technology to transform the entire scope of business operations. They can profoundly combine the digital concept with the enterprise green development strategic planning with vision, including optimization of the production process and green management of the supply chain to the green design of products and other levels, fully utilize the role of digital technology, and achieve the organic unity of enterprise economic gain and environmental gain. And lastly, as the risk control takes it, the epoch of the digital economy is filled with numerous uncertainties and threats and challenges. And the digital background of executives ensures that they are better placed to deal with them. They can benefit from the digital technology, which is highly advanced, to perform real-time monitoring, proper analysis, and effective prevention of different risks experienced by enterprises, in the process making enterprises better placed on the path of green development and reducing the interference and hindrance of risks on implementations of enterprises' green development plans.

In light of the discussion above, the study will focus on data on personal traits of directors, supervisors, and top managers of the publicly traded companies. It makes the professional background of senior managers the starting point and creates a dummy variable to find out the digital background of senior managers. Speaking more precisely, in case the academic major of a particular senior manager contains some keywords, such as "information, intelligence, software, electronics, communication, system, network, automation, wireless, and computer", it will be assumed that this or that senior manager has a digital professional background. Still maintaining that the company itself is a unit of observation, assuming that there is at least one senior manager with the aforementioned digital professional background at the director, supervisor, and senior manager levels of the listed company, the dummy variable will have a coding of 1. Conversely, the value will be 0 in case the directors and supervisors lack such a professional background. Table 10 shows the results in columns (1) and (2). Column (1) indicates that when there are executives with digital backgrounds in the enterprise, the result is significantly positive at the level of 1 %. Column (2) indicates that the findings are not significant in that no managers have a digital background. This finding shows that the digital background of the managers has a substantial and positive role in the building of a link between the digital economy and the green development of businesses.

6.2.4. Digital economy industry

Various segments of the digital economy have different technological aspects, business models, and market structures. The fact that these sectors exhibit a high level of digitalization development predominantly affects the motivation effect of digital technology innovation. Enterprises in industries that are highly digitalized have an opportunity to utilize the further expansion of industrial digitalization to obtain more abundant resources of digital technologies, high-quality digital products and services, and a broad digital infrastructure. All these elements are synergistic in nature, and they lead to the establishment of an enabling environment in which enterprises carry out digital technology innovation activities to enhance the promoting effect of the digital technology

Table 10
Analysis of executives' digital background and industry heterogeneity of the digital economy.

	(1)	(2)	(3)	(4)
	GREEN	GREEN	GREEN	GREEN
Dige	0.085***	0.010	0.025*	0.007
	(2.93)	(1.62)	(1.73)	(1.16)
Size	0.010**	0.004***	0.007***	0.003***
	(2.51)	(5.00)	(4.11)	(3.40)
Lev	−0.022*	0.004	−0.004	0.006*
	(-1.68)	(1.29)	(-0.61)	(1.86)
ROA	−0.016	0.005	0.010	0.002
	(-0.84)	(0.88)	(1.10)	(0.32)
Growth	−0.000	−0.000	0.000	−0.000
	(-0.05)	(-1.17)	(0.60)	(-1.11)
Board	−0.009	−0.007***	−0.009	−0.007***
	(-0.73)	(-2.75)	(-1.39)	(-2.61)
Dual	0.004	−0.001	0.002	−0.002
	(0.96)	(-0.88)	(1.22)	(-1.57)
Top1	−0.007	−0.000	−0.013	0.002
	(-0.21)	(-0.10)	(-1.02)	(0.31)
SOE	−0.006	0.004**	0.005	0.004**
	(-1.02)	(2.18)	(1.26)	(1.98)
Road	0.024**	−0.005**	0.002	−0.004*
	(2.26)	(-2.01)	(0.37)	(-1.70)
Infor	−0.085	−0.010	0.031	−0.020
	(-0.86)	(-0.48)	(0.64)	(-0.94)
Gove	0.213***	−0.006	0.090*	−0.004
	(2.74)	(-0.41)	(1.83)	(-0.28)
_cons	−0.496***	0.027	−0.141*	0.042
	(-3.19)	(0.82)	(-1.75)	(1.21)
N	2055	24,389	4729	21,835
Adj. R^2	0.546	0.489	0.511	0.486

innovation (Tao and Zhan, 2025). First, the digital economy's industries are equipped with superior technologies like cloud computing and artificial intelligence to streamline internal processes and management, which leads to effective resource distribution and energy saving and emission reduction. They are also able to take full advantage of big data analytics in terms of green development by discovering the future green market needs and the technological innovation pathway to accelerate the green technology innovation process. Second, the models of their production and work are relatively traditional in the non-digital economy sectors, as, in the traditional manufacturing and agriculture sectors, they have a weak base of the digital foundation, and their green development process has various challenges and opportunities. Initially, during the early digital economy penetration, these industries might not be experienced and have technological reserves in digital transformation, making the pace of green technology innovation rather sluggish. They will take longer and more resources to adjust to and assimilate digital technologies. The relationship between them will have different results depending on the industry characteristics. The analysis of heterogeneity gives a theoretical foundation for differentiated strategies of digital economy-driven green development of various industries. The current paper affixes the computer, communication, and other electronic equipment manufacturing, software and information technology services, internet and other services and telecommunications, and broadcasting, television, and satellite transmission services as the digital economy industries and carries out the heterogeneity analysis, referring to the "Guidelines to the Industry Classification of Listed Companies" that the CSRC published in 2012. Table 10 presents the estimates in columns (3) and (4). In the digital-economy sector, column (3) indicates that the coefficients are positive and significant at 10 %, whereas column (4) shows that the estimates are insignificant in the non-digital-economy industries. The findings indicate that the positive influence of the digital economy on the green development of enterprises in the digital economy industry is more significant.

6.3. Economic consequences test

As the two driving forces of high-quality development, the digital economy and green development have a profound internal relationship. As the core subject of economic activities, the enterprises' green development process occupies a key position in promoting the overall green development pattern. With its unique advantages, the digital economy is deeply integrated into the process of enterprise green development, thereby rendering the link between the digital economy and green development a topic that the industry and scholars jointly discuss. This study concentrates on verifying the facilitative path of the digital economy on the green development of enterprises, that is, to achieve this goal through green technology innovation and green management innovation. However, whether the digital economy can improve corporate green total factor productivity (GTFP) by guiding enterprises to carry out green development is still a core issue that the business community and many scholars pay close attention to. In line with the endogenous growth theory, economic growth is mainly driven by internal factors of the economic system, and these factors, as endogenous variables, support long-term stable economic growth. The digital economy first drives the green transformation of enterprises through multiple paths. Specifically, during production, digital technology is relied on to optimize the production process, and precise planning is used to reduce resource loss and improve energy utilization efficiency. In the field of technological innovation, we will explore new directions of green technology research and development with the help of big data, artificial intelligence, and other technical means to accelerate the process of technological innovation. In terms of supply chain collaboration, digital platforms should be used to realize green information sharing among upstream and downstream enterprises and strengthen partnership and linkage to build a green supply chain system (Wen et al., 2024). These measures promote enterprises to integrate green concepts into all links of production and operation and practice green actions, providing strong support for the development of GTFP. Drawing on the research of Han et al. (2025), this paper uses the non-radial SBM-ML index to accurately measure corporate GTFP and constructs model (3) to test its economic consequences in depth. The findings appear in column (4) of Table 6, which is significantly positive at the level of 10 %, indicating that the digital economy can ultimately achieve effective improvement of corporate GTFP by promoting enterprises to achieve green development as an important path. The expansion and analysis of this economic consequence can deeply analyze the internal relationship among the digital economy, enterprise green development, and GTFP, and provide an important theoretical basis and practical reference for enterprises to formulate digital promotion and green development strategies.

$$GTFP_{i,t} = \alpha_0 + \alpha_1 GREEN_{i,t} + \alpha_2\, Control_{i,t} + \sum year + \sum ind + \sum province + \varepsilon_{i,t} \tag{3}$$

7. Conclusions and implications

The chapter begins by exploring the inherent reasoning behind the functioning of the digital economy in the green development of corporations and then chooses A-share listed companies in Shanghai and Shenzhen from 2012 to 2023 as the study sample. It is systematic and aims at investigating how the digital economy affects the green development of corporations and their disproportionate performance in different circumstances. At the same time, it concentrates on two aspects, green technology innovation and green management innovation, and thoroughly discusses the particular channels that the digital economy facilitates for corporate green development. The key takeaways from this research are the following: First, the empirical findings demonstrate that the process of digital economy development brings about a strong promoting impact upon enhancing the degree of corporate green development, while this promoting effect is significantly different in various situations. For enterprises with public environmental attention, high level of marketization, digital background of executives and digital economy industries, Second, the green economy can be advanced through the digital economy to enhance green technology innovation as well as green management innovation. Third, the efficacy of the digital economy on corporate green development is increasingly growing; the higher the rate of digital economy development, the more opportunities and potential companies have to access green development, which will positively influence the holistic green development of society. Fourth, the digital economy can incorporate green concepts and technologies into business, which will in turn enhance the enhancement of corporate green total factor productivity when encouraging corporate green development.

Following the research findings identified above, the policy suggestions and the practical pathways are offered. First, sincerely realize the green development concept of "digital empowerment and industrial transformation" to transform the constant engine of the digital economy into the green development of enterprises. At the city level, enhance the establishment of a digital technology innovation system, spur the use of the latest technologies, including 5G, big data, and artificial intelligence, and, concurrently, enhance the profound penetration of the digital economy into the traditional industries, allowing the traditional industries to change and evolve towards green and low-carbon. On the enterprise level, actively respond to the trend of digital economy development, comprehensively implement digital technologies into the business processes of production and product development, and introduce a permanent impetus to green transformation through changing models of traditional production, as well as optimizing methods of resource distribution. Second, as the development of green is highly heterogeneous among enterprises in the digital economy environment, guidance should be classified, and accurate policies should be

developed. More actively encourage the building and enhancement of the systems and avenues of general participation in environmental protection, like the development of environmental complaint hotlines and online complaint platforms, so as to welcome the participation of the masses in environmental supervision and reporting. Meanwhile, further lower administration, enhanced regulation of the market, and enhanced marketization of governmental services to advance them even more. Moreover, it will allow the government to develop a clear digital strategy, enhance the development of technological infrastructure, and enhance the digitalization of enterprise business processes, therefore facilitating the digitalization of enterprises. To conclude, it is crucial to note that the digital economy exerts a leading role in the green development of enterprises and can be a significant lever to use in facilitating green transformation. Simultaneously, the government must complement management and incentive policies and optimize policy control, taking into account the industry standards, enterprise potential, and stakeholder needs to promote enterprises to significant green development.

CRediT authorship contribution statement

Shengnian Wang: Writing – review & editing, Writing – original draft, Visualization, Supervision, Software, Conceptualization. **Qianxing Jiang:** Writing – original draft, Methodology, Conceptualization. **Jinke Wu:** Writing – original draft, Methodology, Investigation.

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Declaration of competing interest

All authors disclosed no relevant relationships.

Data availability

Data will be made available on request.

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